

Point Cloud Compression, Enhancement and Applications: From 3D Perception to Large Models

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Abstract

Point clouds have the strong capability for modeling 3D objects and scenes, which can be widely used in diverse applications and thus generate the burdens of transmission and storage. Efficient compression algorithms have been explored extensively, and research efforts have also been invested to enhancement algorithms. Moreover, the quality of point clouds can influence 3D analysis tasks, e.g., classification, segmentation, detection, and multimodal understanding, etc. Recent 3D multimodal large models can bring better perception optimizations. This tutorial will provide the fundamental knowledge for point cloud compression, enhancement and applications, and place emphasis on the influences of point cloud quality to human and machine perceptions. We will also discuss the progress of international standards and open source projects for point cloud technologies. From this tutorial, audiences are expected to grasp the basic knowledge and recent progress of point cloud technologies, and promote the research developments in both academia and industrial communities.

CCS Concepts

• Information systems → Mobile information processing systems; • Mathematics of computing → Coding theory.

Keywords

Point Cloud Compression, Point Cloud Enhancement, Point Cloud Applications, 3D Human and Machine Perception, Large Models

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1 Context and Objectives

Point cloud data is a new type of data to represent 3D objects and scenes [3], which has broad applications, such as immersive media, autonomous driving, digital heritage protection, robotics, etc. Point cloud can provide high-precision sensing information by the means of 3D coordinates and the corresponding attributes (e.g., color, reflectance, etc.) acquired from RGB-Depth camera and Light Detection And Ranging (LiDAR) equipment. A point cloud can have a large number of points, ranging from thousands to millions or beyond ten millions. For 3D objects, dense point clouds can bring immersive 3D visual experience, while for 3D scenes, sparse point clouds can enhance machine analysis performances. As the fast developments of 3D sensing technologies and applications, efficient point cloud compression methods [6] should be developed. Besides, point cloud enhancement methods should also be elegantly devised to further improve the quality of point clouds.

Point cloud compression can be divided into two categories, including non-learning-based and learning-based methods. Non-learning-based category adopts the paradigm of hybrid structure with typical prediction, transform and entropy modules. Learning-based category fully explores the coding potentials from autoencoder neural network structures. Standardization organizations have promoted the technologies during the past few years. Moving Picture Experts Group (MPEG) group provides two main point cloud compression solutions and standards, including Geometry-based Point Cloud Compression (G-PCC) and Video-based Point Cloud Compression (V-PCC) [16]. It is currently also extending their efforts to develop deep learning-based methods, namely AI-based 3D Graphics Coding (AI-3DGC) [6]. China Audio Video Coding Standard (AVS) group has also just released its first-generation standard [6, 21], and is focusing on AI-based point cloud compression solutions [2, 13, 24–27, 30, 36]. Based on standards, researchers are trying to provide different optimization methods for better coding performances [5]. Nevertheless, performance gains can also be found from non-standard frameworks [12, 14, 15, 17, 18]. Since the recipient of point clouds is either human or machine, point cloud coding methods can be optimized for different applied perceptions.

Point cloud enhancement is another critical aspect to guarantee the quality. Enhancement includes spatial and temporal interpolations, i.e., upsampling [7] and frame interpolation, as well as signal fidelity elevation, i.e., completion [33], compression artifacts removal [8, 29, 34]. Point cloud enhancement can be deemed as pre-processing and post-processing steps of compression task. For both compression and enhancement, quality assessment [1, 19, 20] plays an essential role. Additionally, point clouds are widely applied in machine analysis [9–11, 28, 31, 32] and foundation models. We also

have established open source projects for point cloud technologies, such as OpenPointCloud [4, 35], and others [22, 23].

2 Tutorial Overview

Compression and enhancement are the two main stages to improve point cloud quality, which can have significant impacts on human and machine perceptions. Focusing on these problems, we will also introduce the basic concept of point cloud data and its applications, the quality assessment methods, 3D machine analysis and large models, as well as standardization and open source efforts.

- I. Introduction
 - (1) Concept and Data Acquisition of Point Clouds
 - (2) 3D Human and Machine Perception
- II. Point Cloud Compression
 - (1) Hybrid-based Compression (MPEG V-PCC and G-PCC, AVS PCC, Low Complexity and Rate Control Optimization)
 - (2) AI-based Compression (Point-based, Octree-based, Voxel-based, Large Model-based)
 - (3) Human and Machine Perception-based Compression
- III. Point Cloud Enhancement
 - (1) Spatial and Temporal Enhancement (Upsampling, Completion, Frame Interpolation)
 - (2) Compression Artifacts Removal
- IV. Point Cloud Applications
 - (1) Point Cloud Applications in Machine Analysis (Classification, Segmentation, Detection)
 - (2) Point Cloud Applications in Foundation Models (Pre-training, Multimodal Foundation Models, 3D Embodied AI)
- V. Open Source Projects for Point Clouds
- VI. Conclusion and Discussions

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